



**CRYPTOMATHiC**

# **CUSTOMER STORY: BARCLAYS' QUANTUM-READY TRANSFORMATION WITH CRYPTOMATHIC**



# WHO IS BARCLAYS?

Barclays is a global financial services leader with a legacy spanning over 300 years. Known for its innovation, Barclays has consistently advanced banking technology, from launching the UK's first credit card to installing the world's first ATM. Operating in over 50 countries, Barclays has adopted a proactive stance toward digital transformation, prioritizing both present-day efficiency and future readiness. Cryptographic security has become a cornerstone of its strategy, ensuring scalable and cost-efficient operations while preparing for quantum computing threats.

## IN SUMMARY

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### PROBLEM

Barclays faced escalating cryptographic costs, siloed HSM infrastructure, and the challenge of preparing for quantum computing threats.



### SOLUTION

Cryptomathic's Crypto Service Gateway (CSG) centralized cryptographic management, reduced infrastructure complexity, and provided a scalable, quantum-ready framework.



### RESULTS

Cost savings that freed up budget for post-quantum initiatives, reduced deployment timelines, enhanced cryptographic agility, and future-proof security infrastructure.



### THE CUSTOMER VIEW

"CSG has enabled us to accelerate the delivery of crypto applications while preparing us for the future of quantum-safe cryptography."

Karen Jordan, Director of Cryptography, Barclays Bank.



# CHALLENGES

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Barclays faced growing challenges in managing cryptographic operations:

### 1. ESCALATING COSTS

Managing hundreds of siloed hardware security modules (HSMs) led to rising infrastructure costs and inefficiencies.

### 2. COMPLEXITY AND INEFFICIENCY

Fragmented systems slowed cryptographic updates and operations, creating delays in project delivery and innovation.

### 3. QUANTUM THREATS

With quantum computing poised to disrupt traditional cryptography, Barclays needed to future-proof its infrastructure to secure sensitive operations.

### 4. LIMITED AGILITY

Adapting to new cryptographic standards or integrating post-quantum algorithms was cumbersome, leaving gaps in readiness.

## THE IMPORTANCE OF CRYPTOGRAPHIC AGILITY AND POST-QUANTUM READINESS

Cryptographic agility is essential for banks to adapt to evolving security needs, ensuring resilience against immediate threats while paving the way for future innovations. It involves the ability to update algorithms, switch cryptographic vendors, and maintain strong security postures without disrupting operations. Post-quantum cryptography (PQC) adds another layer of importance, addressing vulnerabilities posed by the computational power of quantum systems.

For Barclays, achieving cryptographic agility addressed present-day inefficiencies and cost burdens, while preparing the organization for quantum-resistant algorithms and standards.



## WHY CRYPTOMATHIC?

Barclays partnered with Cryptomathic to implement the Crypto Service Gateway (CSG), a state-of-the-art cryptographic management solution designed to enhance agility, reduce costs, and prepare for quantum-safe operations. Cryptomathic stood out for its ability to:

### **CENTRALIZE MANAGEMENT**

CSG consolidated Barclays' fragmented cryptographic operations, reducing inefficiencies across 200+ applications globally.

### **DELIVER IMMEDIATE ROI**

By optimizing existing HSM capacity, CSG provided significant cost savings that could be reinvested into post-quantum initiatives.

### **SUPPORT FUTURE STANDARDS**

The modular architecture of CSG enabled seamless integration of new algorithms and standards, ensuring Barclays' readiness for quantum-safe cryptography.

### **ENABLE VENDOR NEUTRALITY**

CSG supported multiple HSM vendors, eliminating vendor lock-in and providing flexibility for future-proofing cryptographic infrastructure.

## THE NATURE OF THE SOLUTION

Cryptomathic's Crypto Service Gateway (CSG) provided Barclays with a comprehensive solution for present and future cryptographic needs:

### **CENTRALIZED CONTROL**

Unified cryptographic operations reduced silos, simplified compliance, and optimized resource utilization.

### **AGILITY IN THE PRESENT:**

CSG allowed Barclays to adapt quickly to changing cryptographic requirements, whether updating algorithms, adjusting key lengths, or integrating new HSMs.



### POST-QUANTUM FLEXIBILITY

Designed to integrate seamlessly with post-quantum algorithms, CSG ensures that Barclays can transition to quantum-safe operations without disruption.

### CRYSTALKEY INTEGRATION

With CSG now part of Cryptomathic's broader CrystalKey offering, Barclays benefits from enhanced lifecycle management and cryptographic agility.

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## RESULTS

Barclays achieved transformative results through its partnership with Cryptomathic:

### 1. REDUCED COSTS

By centralizing operations and optimizing HSM usage, Barclays cut hardware costs significantly, freeing budget to invest in post-quantum initiatives.

### 2. ENHANCED AGILITY

Cryptographic bottlenecks were eliminated, allowing Barclays to reduce deployment timelines from months to weeks.

### 3. FUTURE-READY INFRASTRUCTURE

Barclays is now equipped with a scalable, modular platform that supports post-quantum cryptography and ensures security for decades to come.

### 4. GLOBAL SCALABILITY

The CSG framework operates seamlessly across Barclays' global network, managing cryptographic operations from central hubs in the UK, Singapore, and the United States.

### 5. OPERATIONAL SIMPLIFICATION

Unified cryptographic policies simplified compliance processes and reduced audit costs.



## CONCLUSION

Barclays' adoption of Cryptomathic's Crypto Service Gateway demonstrates how cryptographic agility can drive both immediate and future benefits. By reducing costs, centralizing operations, and preparing for quantum computing, Barclays has solidified its position as a leader in secure and innovative financial services. This proactive investment not only simplifies the path to post-quantum cryptography but also creates value today by enabling faster, more efficient cryptographic management.

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